

CAMS Exam Preparation

Domain D: Tools and Technologies to Fight Financial Crime (20 percent of Exam)

Advanced AML Training Framework

Domain D: 20 percent of CAMS Exam — 90+ Slides — Tools, Technology, Data Quality, AI/ML, Screening

Core Concept

All AML tools are only as effective as the data they receive. Poor data quality = failed controls.

Key Data Quality Dimensions:

- Completeness: All required data fields are populated (no null values)
- Accuracy: Data correctly reflects real-world values
- Timeliness: Data is available when needed (no latency)
- Consistency: Same data has same format across systems
- Validity: Data conforms to defined business rules
- Uniqueness: No duplicate records for same entity

Data Quality Failures on the Exam:

- Silent data feed failures (34 percent of wires missing for 14 months)
- Unlinked transactions (Customer ID = NULL)
- Inconsistent name formats causing screening misses
- Duplicate customer records across systems
- Missing beneficial ownership information

[ALERT] 1 [FOCUS] Exam tests whether you understand that data quality is a **COMPLIANCE** responsibility, not just IT.

[SCENARIO] 34 Percent of International Wires Missing for 14 Months A bank's transaction monitoring system relies on daily data feeds from 17 upstream systems. An audit reveals that for 14 months, the feed from the international wire system silently failed for 34 percent of all wires. Those transactions never entered the monitoring system, never generated alerts, and were never reviewed. Estimated \$340 million in wires were not monitored. The failure was a technical misconfiguration with no error messages. No reconciliation controls existed to detect the gap. The compliance team was unaware because no alerts were triggered.

What the Exam Tests: The critical importance of data reconciliation controls and the danger of silent failures.

How to Analyze:

- 1 34 percent missing data = massive monitoring gap
- 2 No alerts could be generated for transactions the system never received
- 3 The absence of reconciliation controls is a foundational failure
- 4 The bank must remediate AND review missed transactions retrospectively
- 5 SARs must be filed for any suspicious activity identified in missed transactions
- 6 This is a material deficiency regardless of whether the failure was accidental

[TRAP] 1 [CORRECT] Correct Answer: The bank must have controls to verify complete data ingestion. The 14-month gap requires retrospective review and SAR filing. Data quality is a compliance responsibility.

Reconciliation Requirements

Banks must be able to prove that ALL transactions entered the monitoring system.

Required Reconciliation Controls:

- Source system transaction count vs. monitoring system intake count (daily)
- Source system total dollar volume vs. monitoring system total volume (daily)
- Hash total comparisons for data integrity verification
- Exception report for ANY discrepancy (automated)
- Timely resolution of exceptions (24-48 hours maximum)
- Escalation path for unresolved exceptions
- Audit trail of all reconciliation activities

Common Data Quality Issues Detected by Reconciliation:

- Missing transactions (data feed failure - most dangerous)
- Duplicate transactions (processing errors)
- Corrupted data (format issues, encoding problems)
- Delayed data (latency beyond acceptable window)
- Incomplete data (missing required fields like customer ID)
- Incorrect data (wrong values, mapping errors)

Risk of No Reconciliation:

- Cannot demonstrate monitoring completeness to regulators
- Undetected data gaps may exist for months or years
- Criminal activity may be processed without detection
- Bank may face enforcement action for monitoring failure

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[SCENARIO] 12 Percent of Transactions Have NULL Customer ID A bank's transaction monitoring system flags a series of large wire transfers. When the analyst tries to investigate, the system shows the transactions are associated with a "Customer ID: NULL" value. The data issue has been present for 8 months. The bank's IT team explains that when customers open accounts through the mobile app, the unique identifier sometimes fails to populate in the monitoring system. Approximately 12 percent of transactions cannot be linked to any customer record. The compliance team cannot determine which customer conducted which transaction.

What the Exam Tests: The importance of customer identification in transaction monitoring and the risk of orphaned transactions.

How to Analyze:

- 1 Unlinked transactions cannot be investigated properly
- 2 The bank cannot determine customer risk or conduct CDD
- 3 Data quality failure with direct AML impact
- 4 Cannot file accurate SARs without customer identification
- 5 Root cause must be fixed immediately
- 6 Process needed to identify and investigate orphaned transactions retrospectively

[TRAP] 1 [CORRECT] Correct Answer: Missing customer identification data is a material AML deficiency. Fix the root cause and develop a process to identify and investigate orphaned transactions.

Data Quality Key Performance Indicators (KPIs)

Compliance must monitor and report data quality metrics to management and the Board.

Essential Data Quality Metrics:

- Data completeness percentage by source system (target: 99.9 percent or higher)
- Data timeliness (average latency from transaction to monitoring)
- Data accuracy (error rate in critical fields like amount, currency, counterparty)
- Reconciliation exception rate and aging of exceptions
- Number of unlinked transactions (NULL customer ID)
- Number of failed data feeds and recovery time
- Duplicate record rate across systems

Governance Requirements:

- Data quality is SHARED responsibility (IT, Compliance, Business)
- Compliance must DEFINE data requirements for monitoring
- IT must IMPLEMENT controls to meet requirements
- Business must ENSURE source system data quality
- Regular data quality review meetings (weekly or monthly)
- Escalation path for unresolved data issues
- Board reporting on material data gaps

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Core Concept

Digital onboarding tools have transformed KYC, but each technology has specific strengths and weaknesses tested on the exam.

Key Onboarding Technologies:

- E-KYC (Electronic Know Your Customer): Digital verification using government databases
- Digital Identity: Verified digital credentials (e.g., government-issued digital ID)
- Facial Recognition: Comparing selfie to government ID photo
- Liveness Detection: Ensuring the person is physically present (not a photo or video)
- Biometrics: Fingerprint, voice recognition, behavioral biometrics
- Geolocation: Verifying customer's physical location during onboarding
- Document Verification: Scanning and validating passports, driver's licenses

[FOCUS] Exam tests which technology is appropriate for which risk scenario.

E-KYC vs. Traditional KYC

E-KYC uses electronic verification of identity documents and databases rather than physical document inspection.

Capabilities of E-KYC:

- Instant verification of government ID documents
- Cross-reference with credit bureaus and government databases
- Biometric matching (face, fingerprint)
- Liveness detection to prevent spoofing
- Reduced onboarding time (minutes vs. days)
- Improved customer experience
- Audit trail of verification steps

Limitations and Risks:

- Database availability varies by jurisdiction
- May not work for customers without government-issued ID
- Deepfake and presentation attacks (sophisticated fraud)
- Data privacy concerns (biometric data storage)
- Not all jurisdictions recognize digital ID
- May not satisfy regulatory requirements for certain high-risk customers

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[SCENARIO] Bank Uses Facial Recognition for High-Risk Account Opening A bank implements facial recognition for all account openings. A customer uses a high-quality video of another person displayed on a tablet screen in front of their face. The system verifies the face matches the ID document but does not detect that the face is a video recording. The account is opened fraudulently. The bank's compliance officer argues that facial recognition alone is sufficient for identity verification.

What the Exam Tests: The difference between facial recognition and liveness detection.

How to Analyze:

- 1 Facial recognition alone cannot distinguish between a live person and a photo/video
- 2 Liveness detection is REQUIRED to prevent presentation attacks
- 3 Liveness detection uses: eye movement, blinking, head movement, skin texture analysis
- 4 The bank's system was insufficient without liveness detection
- 5 High-risk customers may require additional verification methods

[TRAP] 1 [CORRECT] Correct Answer: Liveness detection is necessary to prevent spoofing. Facial recognition alone is not sufficient for high-assurance verification.

Automated and Manual External Data Sources

Onboarding tools pull data from multiple external sources to verify customer identity and risk.

Key External Data Sources:

- Credit Reference Agencies: Verify address history, credit activity (Experian, Equifax, TransUnion)
- Beneficial Ownership Registers: Corporate UBO information (public registers, commercial databases)
- Government Identity Databases: Passport, driver's license, national ID verification
- Sanctions Lists: UN, OFAC, EU, UK HMT, other national lists
- PEP Lists: Commercially available PEP databases
- Adverse Media Databases: Aggregated news, regulatory actions, court records
- Criminal Records Databases: Where legally permitted
- Watch Lists: Internal fraud databases, industry sharing (where permitted)

Integration Considerations:

- Real-time vs. batch processing (real-time for onboarding)
- API availability and reliability
- Data refresh frequency (daily for sanctions, weekly for PEPs)
- Coverage by jurisdiction (not all sources cover all countries)
- Cost per inquiry vs. subscription models

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Sanctions Screening - Critical Technology

Name screening is a mandatory control for all financial institutions. The exam heavily tests screening methodology.

Key Sanctions Lists:

- UN Security Council Consolidated Sanctions List
- OFAC SDN List (Specially Designated Nationals)
- OFAC Sectoral Sanctions List (SSI)
- EU Consolidated Sanctions List
- UK HM Treasury Sanctions List
- Other national sanctions lists (Canada, Australia, Japan, Switzerland)
- Internal watch lists (bank-specific restricted persons)
- Law enforcement requests (314(a) etc.)

Screening Methodology:

- Batch screening: Periodic screening of entire customer database (daily or weekly)
- Real-time screening: Transaction screening at point of processing
- Fuzzy logic: Name matching with variations, misspellings, transliterations
- Phonetic matching: Soundex, Metaphone for name variations
- Threshold scoring: Score-based matching with configurable thresholds

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[SCENARIO] Sanctions Screening Misses Name Transliteration A payment processor uses exact-string matching for sanctions screening. A wire transfer is initiated by "Muammar Gadaffi." The OFAC SDN list contains "Moammar Qaddafi" with 48 alternative spellings. The exact-string system generates no alert because the name does not match any entry exactly. The wire is processed. An internal review identifies the missed match two weeks later. The compliance officer argues that since the name didn't match exactly, no violation occurred.

What the Exam Tests: The requirement for fuzzy logic or algorithmic name matching in sanctions screening.

How to Analyze:

- 1 Exact-string matching is inadequate for names with multiple transliterations
- 2 OFAC expects institutions to use fuzzy logic to catch name variations
- 3 Arabic, Chinese, Russian names have multiple common spellings
- 4 The missed match is a potential OFAC violation
- 5 The bank must file a Voluntary Self-Disclosure with OFAC
- 6 The screening methodology must be remediated

[TRAP] 1 [CORRECT] Correct Answer: Sanctions screening systems must employ fuzzy logic or algorithmic name matching capable of identifying phonetically similar and transliterated name variants.

Risk-Based Onboarding Segmentation

Different customer types receive different levels of onboarding scrutiny based on risk assessment.

Typical Customer Segments and Onboarding Requirements:

Customer Segment	Onboarding Requirements
Low-risk retail (salary account)	Basic CIP, identity verification, no additional screening
Medium-risk retail (business account)	CIP, beneficial ownership (if entity), sanctions screening
High-risk retail (MSB, NPO)	Enhanced CDD, source of funds, PEP screening, adverse media
Corporate (simple structure)	CIP for entity and UBOs, sanctions screening
Corporate (complex structure)	Full EDD, source of wealth, multi-tier UBO identification
PEP (foreign senior official)	Maximum EDD, senior management approval, source of wealth verification
Correspondent Bank	Wolfsberg CBDDQ, on-site inspection, enhanced monitoring

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Transaction Monitoring Overview

Transaction monitoring is the primary detective control for identifying suspicious activity after onboarding.

Key Components of Transaction Monitoring:

- Rules-based scenarios: Pre-defined logic to flag specific patterns
- Customer segmentation: Different rules for different customer types
- Threshold setting: Values that trigger alerts (e.g., amounts over **\$10,000**)
- Statistical testing: Above Threshold Limit (ATL) / Below Threshold Limit (BTL) testing
- Scenario coverage: Ensuring scenarios detect known typologies
- Alert generation: Output of the monitoring system
- Case management: Workflow for investigating alerts

[FOCUS] Exam tests understanding of false positives, false negatives, and calibration trade-offs.

Two Primary Monitoring Approaches

The exam heavily tests the differences between traditional rules-based and AI/ML-based monitoring.

Rules-Based	Machine Learning
Pre-defined, static rules	Learns patterns from data
Easy to explain and audit	Black box - harder to explain
Low false positives if well-tuned	Can have high false positives
Misses new or evolving typologies	Adapts to new patterns
No model drift	Subject to model drift
Simple implementation	Complex implementation
Low computational requirements	High computational requirements
Requires manual rule updates	Can retrain automatically

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The Core Monitoring Trade-Off

Every AML monitoring system faces this fundamental calibration decision.

Term	Definition	Risk
False Positive	Legitimate activity triggers an alert	Wasted analyst time; operational inefficiency; analyst fatigue
False Negative	Suspicious activity generates NO alert	MISSED criminal activity; direct AML deficiency; enforcement risk

[ALERT] 1 [I] Regulator Priority: Detection effectiveness $\downarrow$ operational efficiency. [FOCUS]

Exam tests scenarios where directors raise thresholds to reduce costs, increasing false negatives.

[SCENARIO] Director Raises All Thresholds by 250 Percent A bank generates 8,000 alerts per month. The compliance director reviews metrics showing that 96 percent of alerts are closed as false positives after manual review. Under pressure to reduce operational costs, the director orders the analytics team to raise all threshold values by 250 percent across all monitoring scenarios. The team leader warns that this will significantly increase false negatives. The director replies: "We cannot afford to review 8,000 alerts per month. A 250 percent threshold increase will reduce volume to a manageable level. Any missed activity is acceptable as long as we still file some SARs."

What the Exam Tests: Over-filtering creates unacceptable false negative risk.

How to Analyze:

- 1 250 percent threshold increase is extreme and arbitrary
- 2 No data analysis was conducted to determine appropriate thresholds
- 3 Many suspicious transactions will fall below new thresholds
- 4 The director is prioritizing cost over effectiveness
- 5 Regulators will view this as a material deficiency
- 6 The statement "any missed activity is acceptable" is regulatory non-compliance

[TRAP] 1 [CORRECT] Correct Answer: Raising thresholds must be based on data analysis and validation. Effectiveness cannot be sacrificed for efficiency.

Above Threshold Limit (ATL) and Below Threshold Limit (BTL)

Statistical testing ensures scenarios are properly calibrated and performing as expected.

ATL (Above Threshold Limit) Testing:

- Tests transactions that SHOULD generate alerts based on scenario logic
- Validates that the system correctly identifies high-risk patterns
- Uses known suspicious cases (historical SARs) as test data
- Expected result: 90-100 percent capture rate for ATL cases
- Low capture rate indicates over-filtering or scenario decay

BTL (Below Threshold Limit) Testing:

- Tests transactions that SHOULD NOT generate alerts
- Uses known legitimate transactions as test data
- Expected result: 0-10 percent false positive rate for BTL cases
- High false positive rate indicates under-filtering or overly sensitive scenarios

Validation Requirements:

- ATL and BTL testing should be performed at least annually
- Material changes trigger interim testing
- Results must be documented and presented to management
- Remediation required for out-of-tolerance results

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Ensuring Scenarios Detect Known Money Laundering Typologies

A transaction monitoring system is only effective if its scenarios cover relevant ML/TF typologies.

Required Scenario Coverage by Typology:

- Structuring (Smurfing): Threshold-based alerts for cash deposits just below CTR limit
- Rapid Movement: Funds in and out within short timeframes
- Layering: Multiple transfers through intermediate accounts
- Trade-Based ML: Mismatched invoices, round-dollar trade payments
- Shell Company Activity: Transfers to/from high-risk jurisdictions
- PEP Activity: Large unexplained transfers, complex structures
- NPO/TF Risk: Rapid cash withdrawals, conflict zone activity
- Cybercrime/Ransomware: Payments to known ransomware wallets
- Human Trafficking: Multiple small payments to controller account

Scenario Effectiveness Testing:

- Run historical SAR cases through current scenarios
- Calculate capture rate for each typology
- Low capture rate indicates missing scenarios or over-filtering
- Emerging typologies require new scenario development

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Model Risk Management (MRM) Framework

AML models, including transaction monitoring and sanctions screening, are subject to model risk management requirements.

Key MRM Components:

- Model Inventory: All models documented with owners, risk ratings
- Model Validation: Independent validation before deployment and periodically thereafter
- Performance Monitoring: Ongoing tracking of precision, recall, false positive rate
- Model Drift Detection: Monitoring for performance degradation over time
- Change Management: Documented approval process for model changes
- Model Governance: Policies, procedures, and oversight

Model Validation Requirements:

- Validator must be independent of model development
- Validation must assess conceptual soundness, data quality, outcomes
- Validation report must document findings and recommendations
- Management must respond to validation findings

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[SCENARIO] ML Model Performance Degrades Over 3.5 Years A bank deployed a machine learning transaction monitoring model in January 2022. The validation report at deployment showed 72 percent precision and 81 percent recall. In June 2025, a validation review finds precision has dropped to 51 percent and recall to 62 percent. During this period, the bank introduced three new payment products, customer transaction volumes increased 180 percent, and criminal typologies shifted from cash structuring to cryptocurrency layering. The model was never recalibrated or retrained after deployment. The compliance director states: "The model was validated at deployment and is still operational."

What the Exam Tests: Model drift requires regular recalibration. Models degrade over time.

How to Analyze:

- 1 Models degrade when customer behavior, products, or typologies change
- 2 3.5 years without recalibration is a significant gap
- 3 The bank failed to establish model monitoring and refresh cadence
- 4 Annual recalibration or trigger-based retraining is required
- 5 The performance degradation is material and requires immediate remediation

[TRAP] 1 [CORRECT] Correct Answer: The bank must establish model monitoring and refresh processes. Significant performance degradation requires recalibration or model replacement.

[SCENARIO] Bank Deploys Deep Neural Network with No Explainability A large international bank deploys a deep neural network transaction monitoring model that achieves 89 percent recall and only 3.2 percent false positives - significantly outperforming the prior rules-based system. However, the DNN model is a "black box": compliance analysts cannot determine why a specific transaction was flagged or which features contributed most to the alert. The external auditor rates model governance as "deficient" because analysts cannot explain alert rationales in investigation narratives. The Chief Data Scientist argues that high accuracy justifies the lack of explainability.

What the Exam Tests: Explainability is a regulatory requirement for AML models.

How to Analyze:

- 1 Analysts must understand WHY a transaction was flagged
- 2 SARs based on unexplained model outputs may not withstand legal scrutiny
- 3 Model cannot be validated for bias or errors without explainability
- 4 Regulators require model governance, not just accuracy
- 5 High accuracy does NOT substitute for explainability

[TRAP] 1 [CORRECT] Correct Answer: Model explainability enables effective investigations, model validation, regulatory auditability, and customer dispute resolution. An inexplicable model creates unmanaged risk regardless of accuracy metrics.

[SCENARIO] Bank Uses Unsupervised Model with 94 Percent False Positive Rate A bank deploys an unsupervised machine learning model designed to detect anomalous transaction patterns without prior labeling of suspicious activity. The model identifies clusters of transactions that deviate from historical customer behavior and generates alerts for "unusual activity." The model's developers cannot explain why specific transactions are flagged. The model has a 94 percent false positive rate, but the bank continues to use it because it has detected two complex layering schemes that rules-based scenarios missed. The compliance committee approves the model without independent validation because it is "supplemental."

What the Exam Tests: Unsupervised models still require validation and governance.

How to Analyze:

- 1 94 percent false positive rate is a significant operational burden
- 2 High false positive volume may overwhelm analysts and obscure meaningful alerts
- 3 Independent validation is required regardless of "supplemental" status
- 4 The model's lack of explainability makes validation difficult
- 5 Difficulty does NOT waive the validation requirement

[TRAP] 1 [CORRECT] Correct Answer: Independent validation is required regardless of whether the model is supplemental or primary. The bank must weigh incremental detection value against operational cost.

Ongoing Customer Due Diligence

CDD is not a one-time event. Customer information and risk profiles must be kept current.

Periodic Refresh (Traditional Approach):

- Scheduled reviews based on risk tier (e.g., high risk annually, medium risk every 3 years)
- Customer must provide updated documentation
- Labor-intensive and reactive
- Risk of missing changes between reviews

Perpetual KYC (Continuous Monitoring):

- Continuous monitoring of customer data and transactions
- Automated alerts when customer information changes
- Real-time adverse media screening
- Trigger-based refresh when risk indicators appear
- AI/ML can prioritize which customers need immediate review

Trigger Events Requiring Unscheduled Refresh:

- Change in ownership structure (new UBOs)
- Change in business activity or geography
- Adverse media findings
- PEP designation change
- Unusual transaction patterns
- Law enforcement inquiry

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Risk-Based Prioritization for KYC Refresh

AI/ML can help prioritize which customers need immediate refresh vs. those that can wait.

Factors for AI-Based Prioritization:

- Time since last refresh (older reviews prioritized)
- Risk tier (high-risk customers prioritized)
- Transaction activity volume and velocity
- Changes in customer behavior patterns
- Adverse media hits
- Sanctions or PEP list changes
- Geographic risk changes (e.g., country added to Grey List)
- Product usage changes (e.g., customer started using high-risk products)

Benefits of AI Prioritization:

- Resources focused on highest-risk customers
- Reduced manual effort for low-risk customers
- Faster identification of risk changes
- Ability to refresh thousands of customers based on risk scoring

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Adverse Media as a Risk Indicator

Adverse media screening identifies negative information about customers from public sources.

Sources of Adverse Media:

- Government sanctions lists (OFAC, UN, EU, UK HMT)
- Regulatory enforcement actions (SEC, FinCEN, FCA)
- Court records and criminal proceedings
- Investigative journalism (ICIJ, OCCRP)
- Reputable news media (Reuters, Bloomberg, FT)
- Government procurement records (corruption indicators)
- Law enforcement wanted lists (Interpol Red Notices)
- Professional disciplinary actions

What is NOT Credible Adverse Media:

- Unverified social media posts
- Anonymous blog posts
- Unmoderated online forums
- Competitor allegations without evidence

[FOCUS] Exam tests distinction between credible and non-credible sources.

[SCENARIO] CEO Named in Corruption Inquiry - No Conviction A bank's EDD team discovers through OSINT that a corporate customer's CEO was named (not charged, not convicted) in a foreign parliamentary corruption inquiry 18 months ago. The customer's primary counterparty's director appears on an Interpol Red Notice. The compliance officer argues that because no conviction has occurred, the adverse media is inconclusive and no action is required.

What the Exam Tests: Reasonable suspicion, not conviction, is the standard for AML action.

How to Analyze:

- ❶ OSINT findings constitute credible adverse media
- ❷ Conviction is NOT required for AML action
- ❸ Reasonable suspicion of ML/TF risk is the standard
- ❹ Findings require escalation and potential EDD
- ❺ Interpol Red Notice alone may justify SAR filing

[TRAP] 1 [CORRECT] Correct Answer: OSINT findings do not require a conviction to be actionable. Reasonable suspicion, not proof of guilt, is the standard for AML risk assessment.

Network Analysis for Financial Crime Detection

Network analysis tools identify relationships between entities that may indicate criminal networks.

What Network Analysis Detects:

- Common ownership across multiple accounts
- Shared addresses, phone numbers, or IP addresses
- Circular transactions between related parties
- Money mule networks (layered transfers)
- Trade-based ML networks (common counterparties)
- Shell company networks (shared directors, registered agents)
- Human trafficking networks (controller accounts)
- Terrorist financing networks (donor clusters)

Visualization Capabilities:

- Node-link diagrams showing entity relationships
- Transaction flow mapping
- Time-series analysis of network evolution
- Centrality analysis (identifying key players)
- Community detection (identifying clusters)

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[SCENARIO] Network Analysis Reveals Money Mule Ring A bank's network analysis tool identifies a cluster of 23 accounts with the following characteristics: all accounts were opened within a 48-hour window; all accounts received small P2P transfers (200-500 dollars) from over 100 different external senders; all accounts then aggregated funds and wired to the same offshore shell company; the accounts share common IP addresses (VPN through secrecy haven) and use the same mobile phone number for verification. Individual transaction monitoring did not flag any single account as suspicious because each account's activity appeared isolated.

What the Exam Tests: Network analysis detects patterns that individual account monitoring misses.

How to Analyze:

- ❶ Individual account monitoring cannot see cross-account patterns
- ❷ Network analysis reveals the coordinated activity across 23 accounts
- ❸ The pattern is classic money mule network
- ❹ The bank should file SARs on all 23 accounts
- ❺ The shared IP addresses and phone numbers are critical indicators

[TRAP] 1 [CORRECT] Correct Answer: Network analysis is essential for detecting coordinated criminal networks that individual account monitoring cannot identify. The pattern requires SAR filing.

Screening Payment Messages

Payment messages contain structured and unstructured data that must be screened for sanctions and AML.

Key Payment Message Types:

- SWIFT MT messages (international wire transfers)
- ISO 20022 (successor to SWIFT MT, richer data)
- ACH (domestic automated clearing house)
- Fedwire (US real-time gross settlement)
- CHIPS (US clearing house interbank payments)
- SEPA (eurozone payments)

SWIFT MT Fields Requiring Screening:

- Field 50: Ordering Customer (originator name and address)
- Field 52: Ordering Institution (originating bank)
- Field 56: Intermediary Institution (correspondent bank)
- Field 57: Account With Institution (beneficiary bank)
- Field 59: Beneficiary Customer (beneficiary name and address)
- Field 70: Remittance Information (payment purpose - free text)

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Screening Blockchain Transactions

Virtual Asset Service Providers (VASPs) must screen blockchain transactions for sanctions and illicit activity.

Blockchain Screening Capabilities:

- Address screening against OFAC SDN list (crypto wallet addresses)
- Transaction tracing through blockchain explorers
- Risk scoring of counterparty wallets
- Travel Rule compliance (originator/beneficiary information)
- Mixer/tumbler detection
- Privacy coin conversion detection
- Cross-chain bridge monitoring

Key Blockchain Analytics Tools:

- Chainalysis: Comprehensive blockchain analysis
- Elliptic: Risk scoring and sanctions screening
- CipherTrace: Transaction monitoring and tracing
- TRM Labs: Real-time risk assessment

Risks in Blockchain Transactions:

- Privacy coins (Monero, Zcash) obscure transaction trails
- Mixers and tumblers break the audit trail
- Cross-chain bridges move value between blockchains
- Decentralized exchanges have no KYC

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Travel Rule Requirements for VASPs

Under FATF Recommendation 16 (updated for virtual assets), VASPs must share originator and beneficiary information.

Required Information for Transfers Above Threshold (typically 1,000 USD/EUR):

- Originator's full name
- Originator's account number (wallet address or account reference)
- Originator's address, national ID number, or date of birth (at least one)
- Beneficiary's full name
- Beneficiary's account number (wallet address)

Receiving VASP Obligations:

- Must obtain required information from sending VASP
- Must take action if information not provided (request, delay, enhanced scrutiny)
- Cannot accept transfers without required information
- May need to file SAR if information cannot be obtained

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[SCENARIO] VASP Receives Transfer with No Originator Information A registered VASP receives a transfer of 4.2 Bitcoin (approximately \$180,000) from another VASP with no originator or beneficiary information attached. The receiving VASP's compliance officer argues that because the customer has already passed KYC, the missing originator data is the sending VASP's problem. The customer states the funds are a personal Bitcoin investment.

What the Exam Tests: Receiving VASP obligations under the Travel Rule.

How to Analyze:

- 1 The receiving VASP has an obligation to obtain required information
- 2 Missing information requires follow-up action
- 3 Actions may include: requesting information from sending VASP, delaying transaction, enhanced scrutiny
- 4 SAR filing may be required if information cannot be obtained
- 5 Customer's KYC at receiving VASP does NOT substitute for originator data

[TRAP] 1 [CORRECT] Correct Answer: The receiving VASP has an obligation to obtain originator and beneficiary information. If not provided, the receiving VASP must take follow-up action, including enhanced scrutiny and potential SAR filing.

Tools to Support AML Investigators

Modern investigation platforms automate many aspects of case management and analysis.

Key Investigation Tool Capabilities:

- Case Management: Workflow for alert review, investigation, SAR filing
- Link Analysis: Visualizing relationships between entities
- Transaction Visualization: Graphical display of fund flows
- Timeline Analysis: Chronological view of customer activity
- Document Management: Storage of investigation notes and evidence
- SAR Generation: Automated population of SAR forms from case data
- Audit Trail: Complete record of investigation actions
- Collaboration Tools: Notes, tasks, assignments for team members

Automation Opportunities:

- Automated data aggregation from multiple sources
- Automated sanctions and PEP screening
- Automated adverse media ingestion
- Automated SAR form population
- Automated closure of low-risk alerts (with validation)
- Automated assignment of alerts to investigators (round-robin or skills-based)

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OSINT for AML Investigations

Open source intelligence uses publicly available information to corroborate or refute suspicion.

Credible OSINT Sources for AML:

- Official government gazettes and sanctions lists
- Corporate registries and beneficial ownership databases
- Court records and legal filings
- Reputable investigative journalism (ICIJ, OCCRP, Bellingcat)
- Regulatory enforcement databases (SEC, FinCEN, FCA)
- Law enforcement wanted lists (Interpol, FBI, Europol)
- Professional licensing boards (disciplinary actions)
- Business information databases (Dun Bradstreet, LexisNexis)

Sources NOT Credible for AML:

- Unverified social media posts
- Anonymous blogs or forums
- Competitor allegations without evidence
- Unmoderated comment sections
- Personal websites with no attribution

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Navigating AML and Privacy Requirements

AML obligations require data collection and sharing; privacy laws restrict it. Technology helps balance these.

Privacy-Enhancing Technologies (PETs) for AML:

- Data Minimization: Collect only what is necessary for AML purposes
- Anonymization: Remove personal identifiers where not needed
- Pseudonymization: Replace identifiers with tokens
- Encryption: Protect data at rest and in transit
- Access Controls: Limit who can view sensitive data
- Audit Logs: Track all access to AML data
- Secure Data Sharing: Encrypted channels for 314(b) and FIU reporting

GDPR-Compliant AML Practices:

- Legal basis for processing: AML legal obligation (Article 6)
- Data retention: 5-year minimum for AML records (Article 17 exception)
- Data subject rights: Right to erasure does NOT apply to AML records
- Data transfers: Allowable for law enforcement cooperation
- Privacy notices: Customers must be informed of AML data processing

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[SCENARIO] Bank Receives Right to Erasure Request for SAR Customer A European bank receives a GDPR "right to erasure" request from a customer demanding deletion of all personal data, including KYC documents and transaction records. The customer was the subject of a SAR filed 18 months ago. The Data Privacy Officer argues GDPR requires deletion. The AML Officer argues AML recordkeeping requires retention (minimum 5 years).

What the Exam Tests: GDPR exception for legal obligations (AML retention).

How to Analyze:

- ① GDPR Article 17(3) exempts data retention required by law
- ② AML recordkeeping (minimum 5 years) is a legal obligation
- ③ The bank can lawfully refuse to delete AML-required records
- ④ SAR confidentiality also prohibits acknowledging the SAR to the customer

[TRAP] 1 [CORRECT] Correct Answer: AML retention requirements are a legal exception to GDPR erasure rights. The bank may refuse deletion for records it is required to retain.

Tool Selection Criteria

Choosing the right AML technology requires evaluating multiple factors.

Key Selection Criteria:

- Risk Profile: Does the tool address the institution's specific ML/TF risks?
- Regulatory Requirements: Does the tool meet regulatory expectations?
- Integration Complexity: Can the tool integrate with existing systems?
- Data Requirements: What data does the tool need? Is it available?
- Scalability: Can the tool handle current and projected transaction volume?
- Performance: Alert generation speed, false positive rate, recall rate
- Explainability: Can the tool explain why it flagged transactions?
- Vendor Reputation: Track record with other financial institutions
- Regulatory References: Has the vendor been examined by regulators?
- Cost: Licensing fees, implementation, maintenance, training
- Support: Vendor support quality and responsiveness

[ALERT] 1

Integrating AML Tools with Existing Systems

Successful AML technology requires seamless integration with core banking and other systems.

Systems That Must Integrate with AML Tools:

- Core Banking System: Customer data, account balances, transaction history
- Payment Systems: Wire transfers, ACH, credit card transactions
- Customer Relationship Management (CRM): Customer contact information
- Sanctions Screening: Real-time and batch screening
- Case Management: Investigation workflow, SAR filing
- Data Warehouse: Historical data for model training and validation

Integration Challenges:

- Data format mismatches (different systems use different formats)
- Data latency (delays between transaction and monitoring)
- API availability (some systems lack modern APIs)
- Data quality (poor data in source systems = poor monitoring)
- Version compatibility (upgrades may break integrations)

Best Practices:

- Conduct proof of concept before full implementation
- Test data feeds thoroughly before go-live
- Establish reconciliation between source and target
- Document integration architecture and data flows

Balancing Effectiveness and Efficiency

AML controls must be effective first; efficiency is a secondary consideration.

Prioritizing Resources Based on Risk:

- Highest-risk customers, products, and geographies receive most resources
- Low-risk areas receive lighter touch (Simplified Due Diligence where permitted)
- Risk assessment drives resource allocation
- Regular review of risk assessment ensures resources remain aligned

Measuring Operational Effectiveness:

- False negative rate (missed suspicious activity) - should be low
- SAR quality (actionable intelligence for law enforcement)
- Investigation timeliness (alerts reviewed within target timeframes)
- Alert-to-SAR conversion rate (meaningful alerts vs. noise)
- Audit findings (fewer findings indicate better controls)

Measuring Operational Efficiency:

- Cost per alert reviewed
- Time per investigation
- False positive rate (higher = less efficient)
- Analyst utilization rate
- SAR filing rate relative to peer institutions

[ALERT] 1

Using Risk Assessment to Prioritize Investment

The risk-based approach means investing most heavily where risk is highest.

High-Risk Areas Requiring More Investment:

- Correspondent banking (highest risk, highest investment)
- Private banking and wealth management (PEP risk)
- Trade finance (TBML vulnerability)
- Virtual assets (emerging typologies)
- Cross-border payments (jurisdictional risk)
- Cash-intensive businesses (structuring risk)

Lower-Risk Areas for Lighter Investment:

- Domestic retail payroll accounts
- Low-balance consumer accounts
- Government benefit disbursement accounts
- Regulated public companies (subject to disclosure)

Documentation Requirements:

- Risk assessment must document resource allocation decisions
- Regulators will review whether resources match risk
- Under-investment in high-risk areas is a deficiency

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Balancing Friction and Risk

Effective AML controls must balance regulatory requirements with customer experience. Too much friction drives customers away or to competitors.

Low-Friction (Good CX) Techniques:

- Digital onboarding with real-time verification (seconds, not days)
- Risk-based friction: EDD only for high-risk customers, SDD for low-risk
- Transparent communication: Explain why information is needed
- Biometric authentication replaces passwords for login
- Mobile-first document upload (scan passport/driver's license)

High-Friction (Bad CX) Red Flags (Over-Controlling):

- Requiring in-person branch visit for every account type
- Asking all customers for source-of-wealth documentation
- Repeatedly requesting same documents already provided
- No digital channel for KYC updates

[FOCUS] Exam tests ability to apply risk-based friction: low-risk low friction; high-risk high friction.

External Data Sources Overview

Onboarding and screening tools pull data from multiple external sources to verify identity, risk, and sanctions status.

Key External Data Sources and Their Use:

Source Type	What It Verifies / AML Use
Credit Reference Agencies (Experian, Equifax, TransUnion)	Address history, credit activity, identity verification
Beneficial Ownership Registers (public/commercial)	Corporate UBO identification (e.g., OpenCorporates, local registries)
Government Identity Databases	Passport, driver's license, national ID verification (real-time API)
Sanctions Lists (UN, OFAC, EU, UK HMT)	Mandatory screening for prohibited persons/entities
PEP Lists (commercial databases)	Identify Politically Exposed Persons (WorldCheck, Refinitiv)
Adverse Media Databases (aggregated news/regulatory)	Negative news, enforcement actions, court records
Criminal Records Databases (where legally permitted)	Past convictions relevant to financial crime
Internal Watch Lists	Bank-specific restricted persons, previous SAR subjects

[ALERT] 1

Batch Screening vs. Real-Time Screening

Sanctions screening occurs at two different frequencies, each with specific requirements.

Batch Screening (Periodic):

- Screens entire customer database against updated sanctions lists
- Frequency: Typically daily, weekly, or after each sanctions list update
- Detects customers who became sanctioned *after* onboarding
- High volume of false positives requires fuzzy logic

Real-Time Screening (Transactional):

- Screens each payment/transaction before processing (holds payment if alert)
- Required for wire transfers, crypto transactions, instant payments
- Must complete screening within milliseconds to avoid payment delays

List Management Considerations:

- Sanctions lists change frequently (OFAC updates daily)
- Must maintain audit trail of which list version was used for each screen
- False positive analysis: Document why a match is not a true positive
- Fuzzy logic (not exact matching) is REQUIRED (as covered in D.2.5)

[FOCUS] Exam tests the difference: batch = customer database; real-time = transactions.

Adverse Media as a Risk Indicator (Expanded)

Adverse media screening identifies negative information from public sources using automated tools and AI.

How Adverse Media Tools Work:

- Web scraping and API feeds from thousands of sources (news, regulators, courts)
- Natural Language Processing (NLP) to classify sentiment and extract entities
- Entity resolution: Matching "John Smith" from article to customer "J. Smith"
- Risk scoring: Assign severity score (e.g., fraud = 8/10, minor traffic violation = 1/10)
- Ongoing monitoring: Continuous screening, not just at onboarding

Actionable Findings (No Conviction Required):

- Naming in a corruption inquiry → Escalate for EDD
- Regulatory fine for AML violations → Reassess relationship
- Indictment (even before conviction) → Consider SAR
- Interpol Red Notice → Immediate SAR filing

[I] FATF expects institutions to screen for adverse media continuously, not just at onboarding.

AI Tools to Support AML Investigators

Beyond alert generation, AI/ML can significantly enhance the investigation workflow.

AI-Powered Investigation Tools:

- Automated data aggregation: Pulls KYC, transactions, sanctions hits, adverse media into single case file (minutes vs. hours)
- Entity resolution: Connects related accounts, addresses, phone numbers across systems
- Risk scoring of alerts: Prioritizes highest-risk alerts for investigator review
- Automated narrative generation: Drafts SAR narratives from structured case data (analyst reviews and edits)
- Link analysis visualization: Automatically maps relationships between entities
- Predictive case closure: Suggests "likely false positive" based on historical patterns (with human validation)

Key Exam Distinction:

- AI can AUTOMATE data collection and prioritization
- AI should NOT replace investigator judgment for SAR decisions
- Investigators must always review and validate AI suggestions

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Tool Selection Criteria (Expanded)

Choosing the right AML technology requires evaluating multiple factors beyond just price.

Comprehensive Selection Framework:

- Risk Alignment: Does the tool address your specific ML/TF risks (e.g., trade finance, crypto, correspondent banking)?
- Regulatory Acceptance: Has the vendor been examined by regulators? Do peer institutions use it?
- Integration Complexity: APIs available? Data format compatibility with core banking system?
- Performance Metrics: Request vendor's false positive rate, recall rate, precision on independent test data
- Explainability: Can the tool explain WHY it flagged a transaction (critical for AI models)?
- Total Cost of Ownership (TCO): License fees + implementation + integration + training + ongoing maintenance
- Vendor Stability: Financial health, support responsiveness, update frequency
- Scalability: Can it handle 2x or 5x current transaction volume without performance degradation?

[FOCUS] Cheapest tool is rarely the best. Effectiveness and explainability are paramount.

Integration Architecture

A tool is only as good as its data feeds. Poor integration = poor detection.

Systems That Must Integrate:

- Core Banking System: Customer master, account balances, transaction history
- Payment Systems: SWIFT, ACH, Fedwire, SEPA, crypto wallets
- CRM: Customer contact info, relationship manager assignments
- Case Management: Workflow integration (alert → case → SAR)
- Data Lake/Warehouse: Historical data for model training and validation
- Sanctions Lists: Real-time API feeds for list updates

Integration Risk Points (Exam Scenarios):

- Data format mismatches (CSV vs. JSON vs. XML)
- Data latency (transaction at 9:00 AM, monitoring system sees it at 9:00 PM)
- Silent feed failures (no error message, data just missing)
- API rate limiting (vendor caps queries, transactions queued and delayed)

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Additional Topics Now Covered

The following slides have been added to complement your existing materials:

- Customer Experience (CX) / Friction management
- External data sources for KYC (credit agencies, govt ID, etc.)
- Batch screening vs. real-time screening
- List management for sanctions
- Adverse media AI/ML tools
- AI in investigations (automation support)
- Comprehensive tool selection criteria
- Integration considerations (APIs, latency, feed failures)
- Total Cost of Ownership (TCO)
- Vendor due diligence

[I] These slides ensure full syllabus coverage for Domain D (20% of CAMS exam).

- 1 Silent data failure = monitoring gap
- 2 Reconciliation controls prevent undetected data loss
- 3 Fuzzy logic required for sanctions screening
- 4 False negatives worse than false positives
- 5 Model drift requires regular recalibration
- 6 Explainability required for AI models
- 7 E-KYC requires liveness detection
- 8 Network analysis detects money mule rings
- 9 Travel Rule applies to VASPs (both sending and receiving)
- 10 Adverse media does NOT require conviction
- 11 Data quality is compliance responsibility
- 12 Perpetual KYC & periodic refresh
- 13 Trigger-based refresh required for material changes
- 14 ATL/BTL testing validates scenario effectiveness
- 15 Rules-based + ML hybrid is best practice
- 16 Unsupervised models still require validation
- 17 SWIFT fields requiring screening: 50, 52, 56, 57, 59, 70
- 18 Blockchain analytics required for VASPs
- 19 Privacy coins and mixers require EDD
- 20 Receiving VASP cannot ignore missing Travel Rule data
- 21 GDPR has AML retention exception
- 22 OSINT must use credible sources only
- 23 Tool selection must consider integration and scalability
- 24 Effectiveness prioritized over efficiency
- 25 Risk-based approach drives resource allocation
- 26 Model risk management requires independent validation
- 27 Sanctions screening requires fuzzy logic, not exact matching
- 28 Over-filtering increases false negatives
- 29 Scenario coverage must address all typologies
- 30 Investigation tools require audit trails

- 34 percent missing data for 14 months → Data failure
- NULL Customer ID on transactions → Unlinked transactions
- No reconciliation controls → Undetected gaps
- 77 percent false negative rate → Material deficiency
- No recalibration for 3.5 years → Model drift
- Black box AI with no explainability → Governance failure
- Exact-string matching only → Sanctions screening failure
- No liveness detection → Spoofing vulnerability
- No adverse media review → Missed risk indicators
- No network analysis → Missed mule rings
- Missing Travel Rule data accepted → VASP violation
- No ATL/BTL testing → Unvalidated scenarios
- Only annual refresh after material changes → Control gap
- Same onboarding for all customers → RBA violation
- No independent model validation → MRM failure

- Domain D is 20 percent of exam
- Data quality is a COMPLIANCE responsibility, not just IT
- Silent data failures (no error messages) are the most dangerous
- FALSE NEGATIVES are worse than false positives
- Fuzzy logic is REQUIRED for sanctions screening
- ML models require explainability and ongoing validation
- Model drift requires regular recalibration
- Travel Rule applies to BOTH sending and receiving VASPs
- E-KYC without liveness detection can be spoofed
- Adverse media findings do NOT require conviction
- Network analysis detects patterns individual monitoring misses
- GDPR has an AML retention exception (Article 17(3))
- Credible OSINT sources: government, courts, reputable journalism
- Privacy coins and mixers require EDD
- Effectiveness *vs* efficiency (regulator priority)
- Risk-based approach drives resource allocation